

Simplifications of Context-Free Grammars

A Substitution Rule

Equivalent
grammar

$$A \rightarrow a$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc$$

$$B \rightarrow abbA$$

$$B \rightarrow b$$

Substitute B

$$A \rightarrow a$$

$$A \rightarrow aaA$$

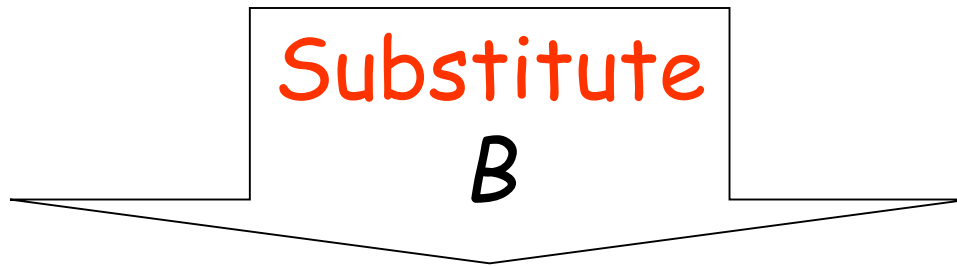
$$A \rightarrow ababbAc$$

$$A \rightarrow abbc$$

In general:

$$A \rightarrow xBz$$

$$B \rightarrow y_1 \mid y_2 \mid \cdots \mid y_n$$



$$A \rightarrow xy_1z \mid xy_2z \mid \cdots \mid xy_nz$$

equivalent
grammar

Useless Productions

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

$$S \rightarrow A$$

$$A \rightarrow aA$$
 Useless Production

Some derivations never terminate...

$$S \Rightarrow A \Rightarrow aA \Rightarrow aaA \Rightarrow \dots \Rightarrow aa \dots aA \Rightarrow \dots$$

Another grammar:

$$S \rightarrow A$$

$$A \rightarrow aA$$

$$A \rightarrow \lambda$$

$$B \rightarrow bA$$

Useless Production

Not reachable from S

In general:

If $S \Rightarrow \dots \Rightarrow xAy \Rightarrow \dots \Rightarrow w$


 $w \in L(G)$

Then variable A is useful

Otherwise, variable A is useless

A production $A \rightarrow x$ is useful
if all its variables are useful

Removing Useless Productions

Example Grammar:

$$S \rightarrow aS \mid A \mid C$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$C \rightarrow aCb$$

First: find all variables that produce strings with only terminals

$$S \rightarrow aS \mid A \mid C$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$C \rightarrow aCb$$

Round 1: $\{A, B\}$



Round 2: $\{A, B, S\}$

Keep only the variables
that produce terminal symbols $\{A, B, S\}$

$$S \rightarrow aS \mid A \mid \cancel{C}$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$\cancel{C \rightarrow aCb}$$



$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

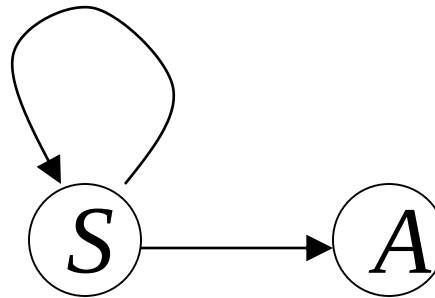
Second: Find all variables
reachable from S

Dependency Graph

$S \rightarrow aS \mid A$

$A \rightarrow a$

$B \rightarrow aa$



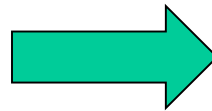
not
reachable

Keep only the variables
reachable from S

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

~~$$B \rightarrow aa$$~~



Final Grammar

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

Nullable Variables

λ – production : $A \rightarrow \lambda$

Nullable Variable: $A \Rightarrow \dots \Rightarrow \lambda$

Removing Nullable Variables

Example Grammar:

$$S \rightarrow aMb$$

$$M \rightarrow aMb$$

$$M \rightarrow \lambda$$

Nullable variable



$S \rightarrow aMb$

$M \rightarrow aMb$

~~$M \rightarrow \lambda$~~

Substitute
 $M \rightarrow \lambda$

Final Grammar

$S \rightarrow aMb$

$S \rightarrow ab$

$M \rightarrow aMb$

$M \rightarrow ab$

Unit-Productions

Unit Production: $A \rightarrow B$

Removing Unit Productions

Observation:

$$A \rightarrow A$$

Is removed immediately

Example Grammar:

$$S \rightarrow aA$$

$$A \rightarrow a$$

$$A \rightarrow B$$

$$B \rightarrow A$$

$$B \rightarrow bb$$

$$S \rightarrow aA$$

$$A \rightarrow a$$

~~$$A \rightarrow B$$~~

$$B \rightarrow A$$

$$B \rightarrow bb$$

Substitute

$$A \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid B$$

$$B \rightarrow bb$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid \cancel{B}$$

$$B \rightarrow bb$$

Remove

$$B \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A$$

$$B \rightarrow bb$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

~~$$B \rightarrow A$$~~

$$B \rightarrow bb$$

Substitute

$$B \rightarrow A$$

$$S \rightarrow aA \mid aB \mid aA$$

$$A \rightarrow a$$

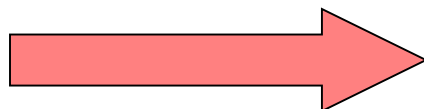
$$B \rightarrow bb$$

Remove repeated productions

$$S \rightarrow aA \mid aB \mid \cancel{aA}$$

$$A \rightarrow a$$

$$B \rightarrow bb$$



Final grammar

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow bb$$

Removing All

Step 1: Remove Nullable Variables

Step 2: Remove Unit-Productions

Step 3: Remove Useless Variables